Bahria University, Islamabad

Department of Software Engineering

Computer Communication & Networking Lab

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Task No:	**Task Wise Marks**		**Documentation Marks**		**Total Marks (20)**
	Assigned	**Obtained**	**Assigned**	**Obtained**	
1					
2					
3					
4					
5					

Comments:

Signature

LAB # 11

Configuring RIP (Routing Information Protocol) routing protocol between two routers

Objective

To enable communication between two hosts that are connected not to a single router but with the different routers

Introduction

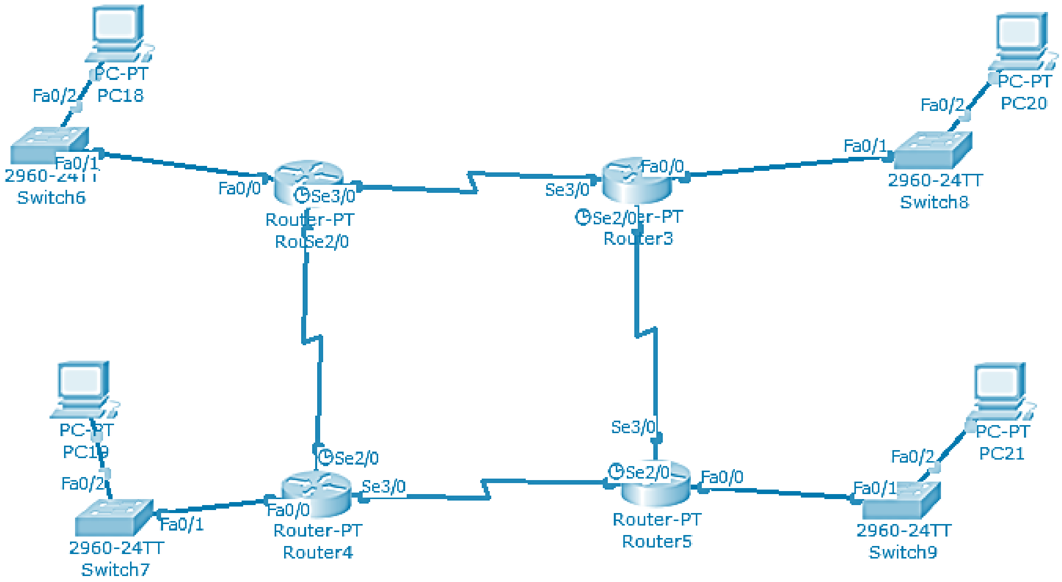
The Routing Information Protocol (RIP) is a relatively old, but still commonly used, interior gateway protocol (IGP) created for use in small, homogeneous networks. It is a ***classical distance-vector*** routing protocol. RIP is documented in RFC 1058.
RIP uses broadcast User Datagram Protocol (UDP) data packets to exchange routing information.

The Cisco IOS software sends routing information updates every 30 seconds; this process is termed **advertising**. If a router does not receive an update from another router for 180 seconds or more, it marks the routes served by the non-updating router as being **unusable**. If there is still no update after 240 seconds, the router *removes all routing table entries* for the no updating router.

The metric that RIP uses to rate the value of different routes is **hop count**. The hop count is the number of routers that can be traversed in a route. A directly connected network has a **metric of zero**; an unreachable network has a **metric of 16**. *This small range of metrics makes RIP an unsuitable routing protocol for large networks*.
If the router has a default network path, RIP advertises a route that links the router to the pseudo network 0.0.0.0. The network 0.0.0.0 does not exist; RIP treats 0.0.0.0 as a network to implement the default routing feature. The Cisco IOS software will advertise the default network if a default was learned by RIP, or if the router has a gateway of last resort and RIP is configured with a default metric.
RIP sends updates to the interfaces in the specified networks. If an interface's network is not specified, it will not be advertised in any RIP update.

Equipment:

1. Four routers
2. Four switches
3. Four PCs



Procedure:

Step1: Assign IP address on all ports of R1, R2, R3 and R4

Assign IP addresses to all ports of Router shown in figure above.

Step2: Configure RIP protocol on all routers

ExampleName#config

ExampleName(config)#router rip

ExampleMame(config-router)#network aa.bb.cc.dd

ExampleName(config-router)# network ee.ff.gg.hh

ExampleName(config-router)#

Step3: Assign IP address to all computer

Assign IP address to computer with same network that assign to their respective port

Step4: confirm connection

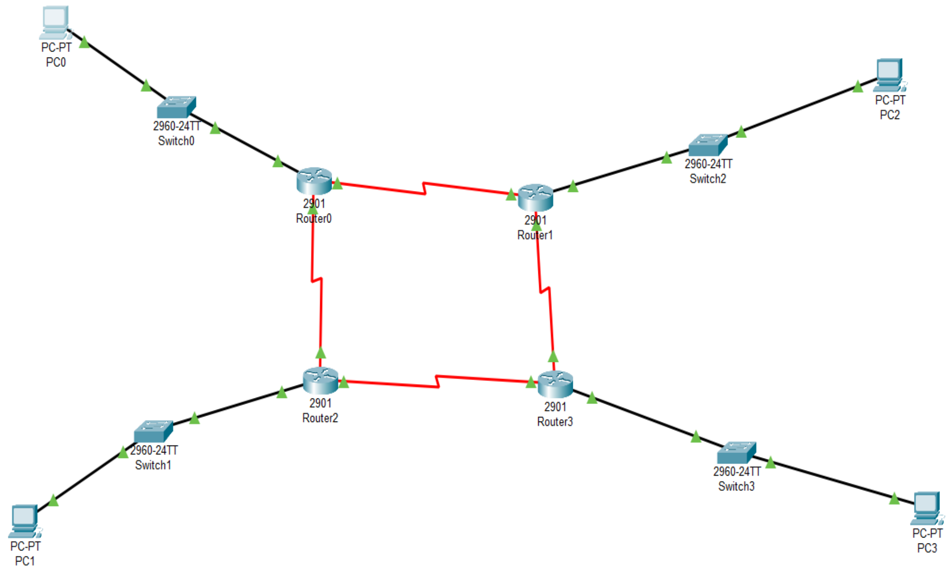
Use ping command to confirm connection of each PC.

Step5: check routing protocol

ExampleName(config)# show ip router

Task:

1. Implement the above-mentioned network using Packet Tracer



PC	IP Address	Subnet Mask	Default Gateway
PC0	192.168.1.10	255.255.255.0	192.168.1.1
PC2	192.168.2.10	255.255.255.0	192.168.2.1
PC1	192.168.3.10	255.255.255.0	192.168.3.1
PC3	192.168.4.10	255.255.255.0	192.168.4.1

LAN Ips

Router	Interface	IP Address	Mask
Router0	Gig0/0	192.168.1.1	255.255.255.0
Router1	Gig0/0	192.168.2.1	255.255.255.0
Router2	Gig0/0	192.168.3.1	255.255.255.0
Router3	Gig0/0	192.168.4.1	255.255.255.0

Serial Ips

Router0 ↔ Router1 Network: **10.0.0.0/30**

Router	Interface	IP
Router0	Se0/0/1	10.0.0.1
Router1	Se0/0/1	10.0.0.2

Router0 ↔ Router2
Network: **10.0.0.4/30**

Router	Interface	IP
Router0	Se0/0/0	10.0.0.5
Router2	Se0/0/0	10.0.0.6

Router1 ↔ Router3

Network: **10.0.0.8/30**

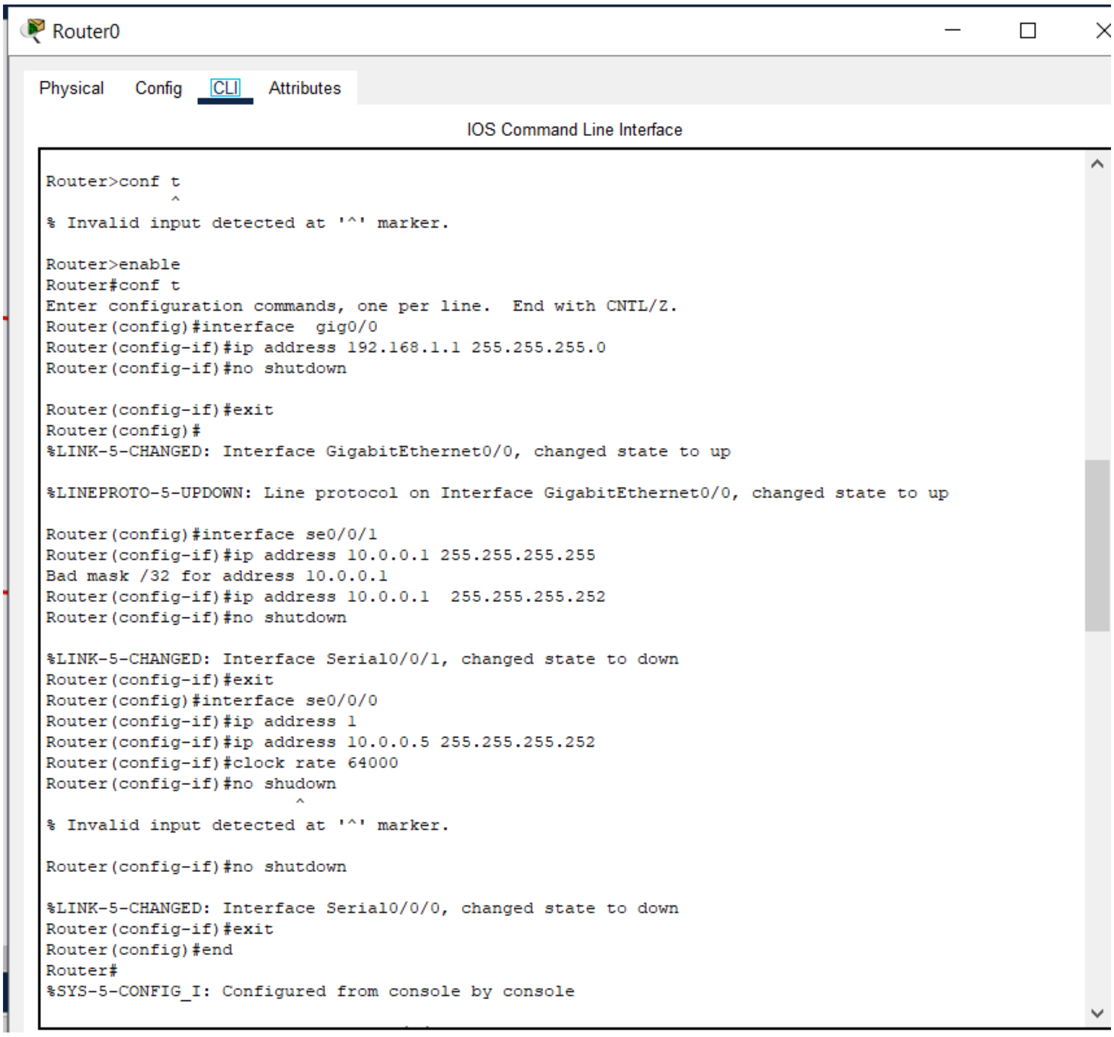
Router	**Interface**	**IP**
Router1	Se0/0/0	10.0.0.9
Router3	Se0/0/1	10.0.0.10

Router2 ↔ Router3

Network: **10.0.0.12/30**

Router	**Interface**	**IP**
Router2	Se0/0/1	10.0.0.13
Router3	Se0/0/0	10.0.0.14

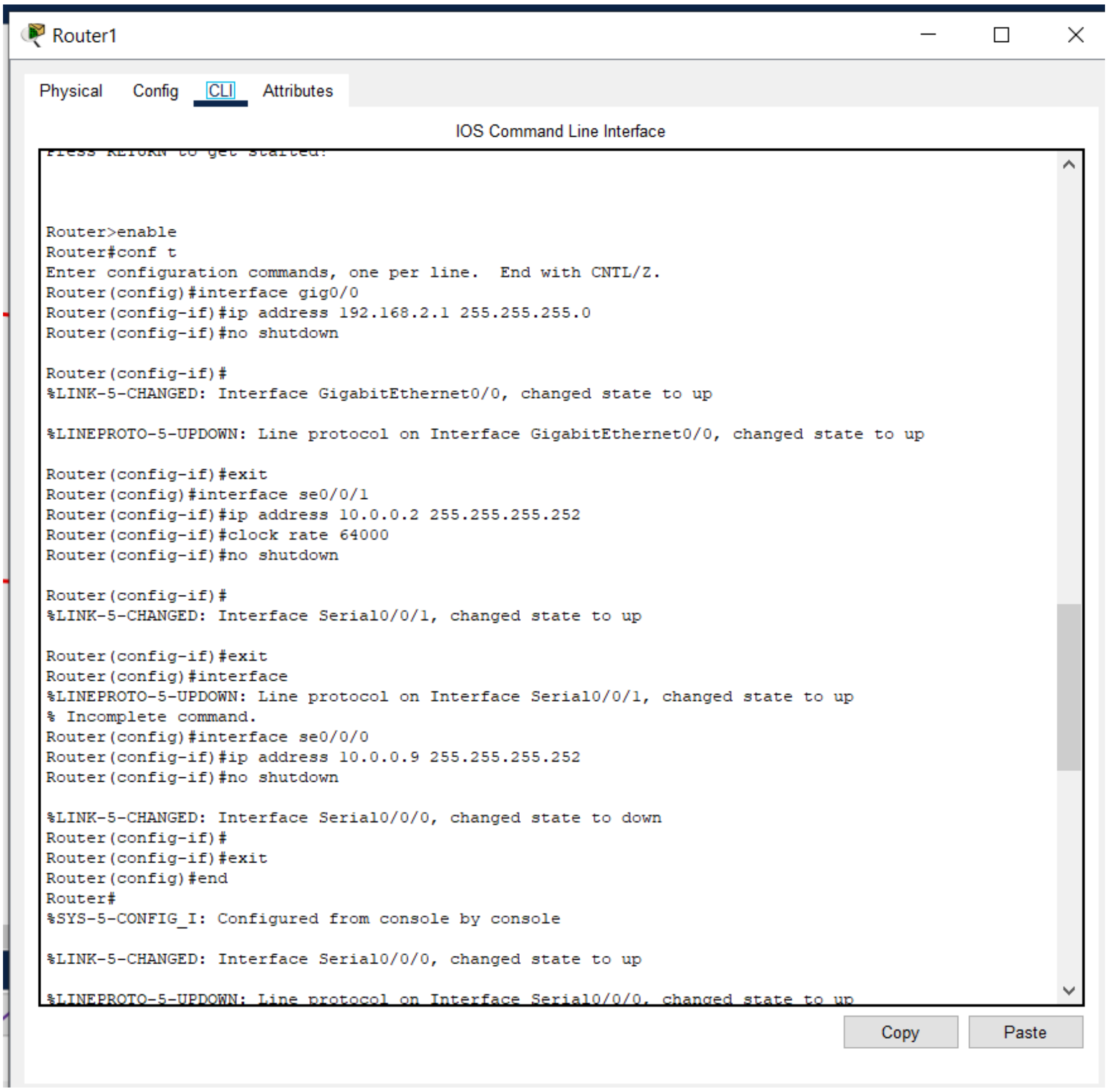
Subnet mask for ALL serial:
255.255.255.252

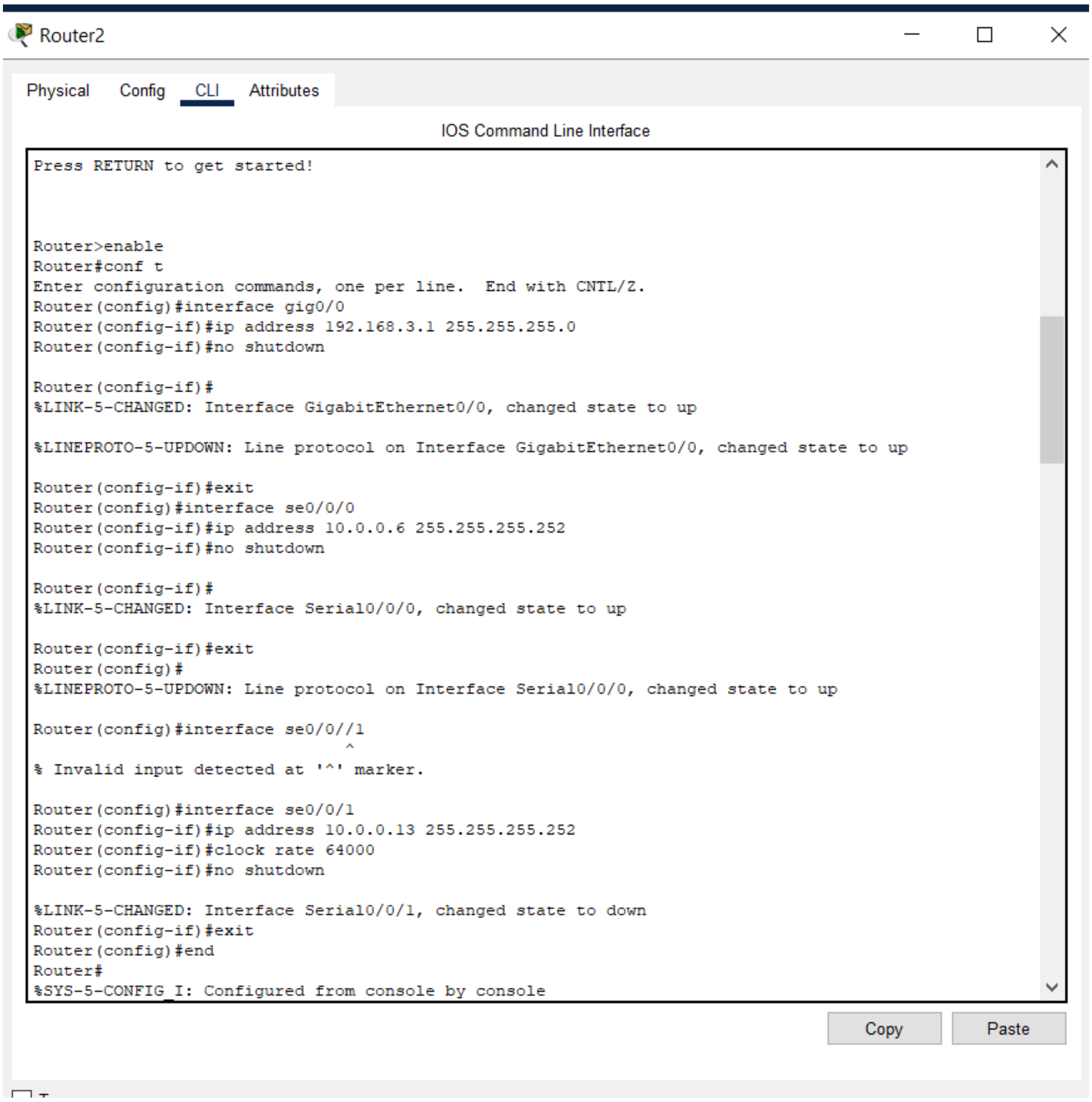

Router0

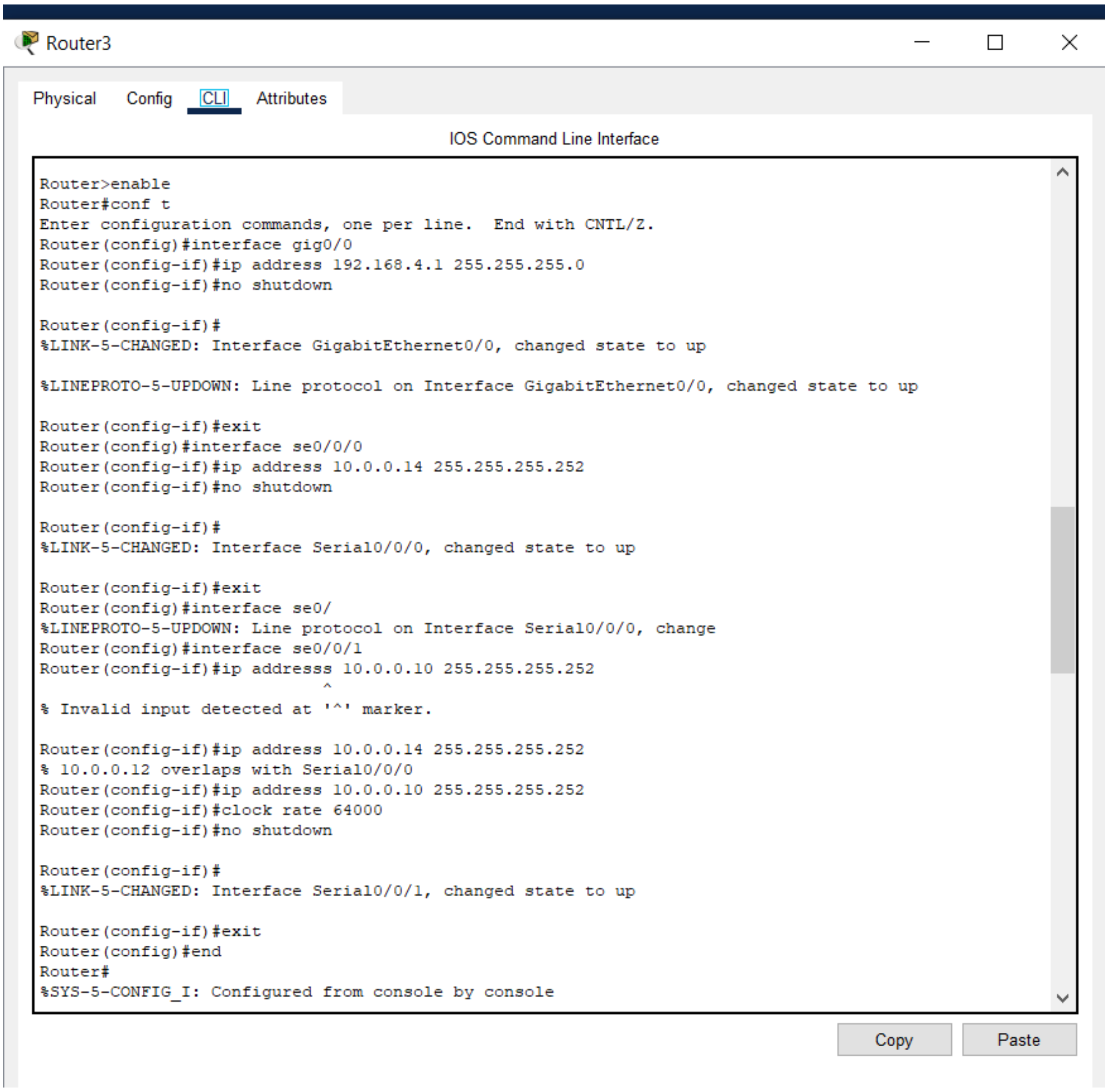
Physical Config CLI Attributes

IOS Command Line Interface

```
Router>conf t
            ^
% Invalid input detected at '^' marker.

Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface  gig0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config)#interface se0/0/1
Router(config-if)#ip address 10.0.0.1 255.255.255.255
Bad mask /32 for address 10.0.0.1
Router(config-if)#ip address 10.0.0.1  255.255.255.252
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down
Router(config-if)#exit
Router(config)#interface se0/0/0
Router(config-if)#ip address 1
Router(config-if)#ip address 10.0.0.5 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shudown
                        ^
% Invalid input detected at '^' marker.

Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```


Router1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#exit
Router(config)#interface se0/0/1
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up
Router(config-if)#exit
Router(config)#interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up
% Incomplete command.
Router(config)#interface se0/0/0
Router(config-if)#ip address 10.0.0.9 255.255.255.252
Router(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
Router(config-if)#
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
Copy
Paste


Router2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started!
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#exit
Router(config)#interface se0/0/0
Router(config-if)#ip address 10.0.0.6 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
Router(config)#interface se0/0//1
^
% Invalid input detected at '^' marker.
Router(config)#interface se0/0/1
Router(config-if)#ip address 10.0.0.13 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Copy
Paste




```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 192.168.4.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface se0/0/0
Router(config-if)#ip address 10.0.0.14 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

Router(config-if)#exit
Router(config)#interface se0/
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, change
Router(config)#interface se0/0/1
Router(config-if)#ip addresss 10.0.0.10 255.255.255.252
                            ^
% Invalid input detected at '^' marker.

Router(config-if)#ip address 10.0.0.14 255.255.255.252
% 10.0.0.12 overlaps with Serial0/0/0
Router(config-if)#ip address 10.0.0.10 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up

Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Configure RIP on all routers

```
Router>enable
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/0/0            10.0.0.5        YES manual up                    up
Serial0/0/1            10.0.0.1        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.1.0
Router(config-router)#network 10.0.0.0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
Router#
```

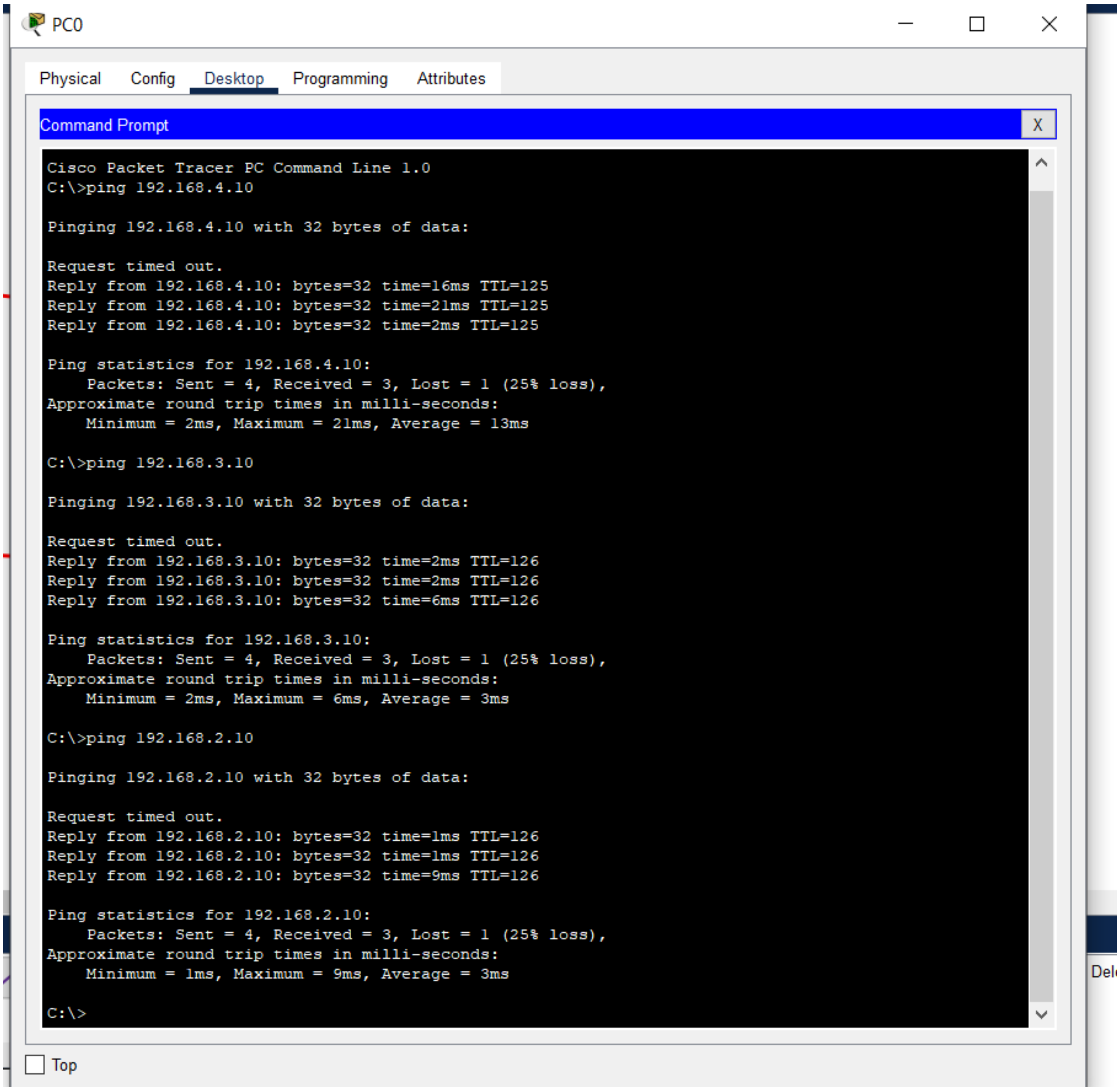
```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.2.1     YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/0/0            10.0.0.9        YES manual up                    up
Serial0/0/1            10.0.0.2        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.2.0
Router(config-router)#network 10.0.0.0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
```

Copy Paste

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.3.1     YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/0/0            10.0.0.6        YES manual up                    up
Serial0/0/1            10.0.0.13       YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router#router rip
               ^
% Invalid input detected at '^' marker.

Router#
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.3.0
Router(config-router)#network 10.0.0.0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
Router#
```

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.4.1     YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/0/0            10.0.0.14       YES manual up                    up
Serial0/0/1            10.0.0.10       YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router#enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.4.0
Router(config-router)#network 10.0.0.0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

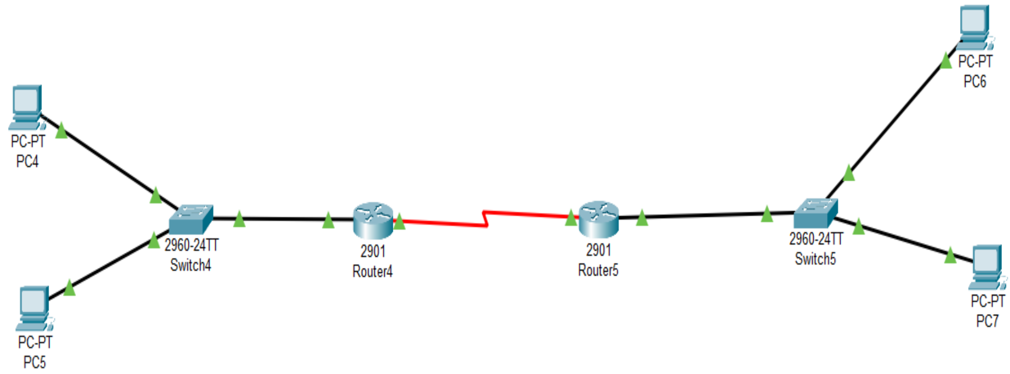
PING:



Connectivity Verification and Results

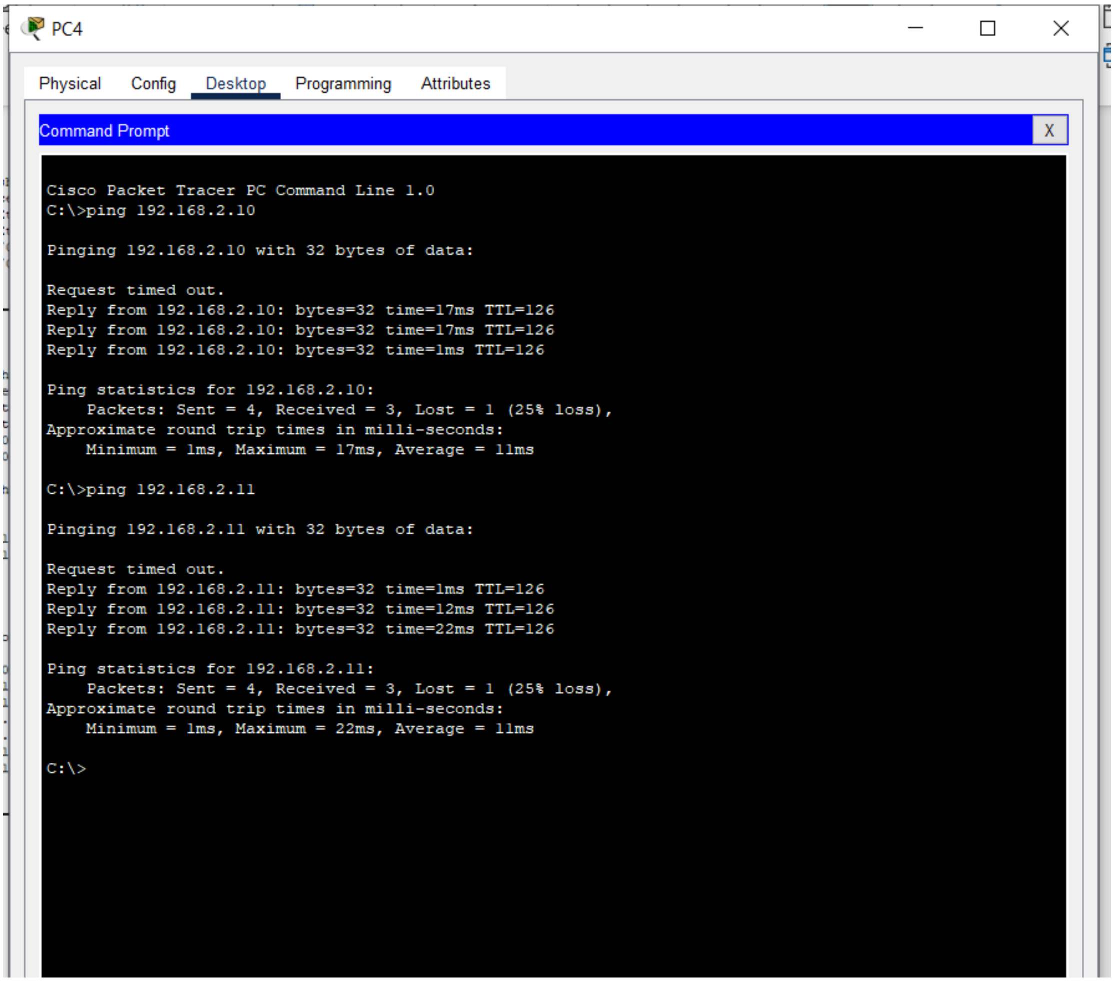
To check whether the network was working correctly, ping commands were used to test communication between PCs on different networks. From PC0, ping requests were sent to PC1, PC2, and PC3, which are connected through different routers. In each case, the first ping request timed out, but the remaining requests were successful. This initial timeout happened because the router needed time to learn the MAC addresses and routing information. After that, the packets were transmitted successfully. The successful replies show that RIP routing is functioning properly and data is being forwarded across multiple routers. This confirms that the network configuration is correct and the devices can communicate with each other.

2.Implement above network using 2 routers and four PCs, two on each side on Packet Tracer.




```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/0/0            10.0.0.1        YES manual up                    up
Serial0/0/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.2.1     YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Serial0/0/0            10.0.0.2        YES manual up                    up
Serial0/0/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
L       10.0.0.2/32 is directly connected, Serial0/0/0
R    192.168.1.0/24 [120/1] via 10.0.0.1, 00:00:08, Serial0/0/0
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/0
L       192.168.2.1/32 is directly connected, GigabitEthernet0/0

Router#
```



IP Addressing

Device	IP Address	Subnet	Gateway
Router4 Gig0/0	192.168.1.1	255.255.255.0	—
PC4	192.168.1.10	255.255.255.0	192.168.1.1
PC5	192.168.1.11	255.255.255.0	192.168.1.1

Router5 : Gig0/0	192.168.2.1	255.255.255.0	—
PC6	192.168.2.10	255.255.255.0	192.168.2.1
PC7	192.168.2.11	255.255.255.0	192.168.2.1

Serial Link (Between Routers)

Network: **10.0.0.0/30**

Router	Interface	IP
Router0	Se0/0/0	10.0.0.1
Router1	Se0/0/0	10.0.0.2

Ping Verification

To verify network connectivity, ping tests were performed between PCs located on different networks. From PC0, ping requests were sent to PCs on the opposite side of the network. In each test, the first ping request timed out, while the remaining requests were successful. This initial timeout occurred due to ARP resolution and route discovery by the routers. After this, ICMP packets were delivered successfully. The successful ping replies confirmed that the routers were correctly forwarding packets and that RIP routing was functioning properly across the network.

Conclusion for LAB:

In this lab, the network was successfully implemented using routers, switches, and PCs in Cisco Packet Tracer. All devices were properly connected and assigned correct IP addresses. Dynamic routing using RIP was configured, which allowed routers to automatically share routing information. Initial connectivity issues were resolved by correctly configuring interfaces and routing. Successful ping tests between PCs on different networks confirmed that the configuration was correct. Through this lab, I was able to understand IP addressing, dynamic routing, and how routers communicate in a multi-network environment.